

COMPARATIVE ANALYSIS OF DEEP LEARNING ARCHITECTURES FOR PNEUMONIA DETECTION IN CHEST X-RAYS

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Course: Applied Machine Learning (Advance)

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GitHub Project link: <https://github.com/hasanmirza72/Pneumonia-Deep-Learning-Prediction-AML-Advance>

1. Summary

This project presents a rigorous comparative evaluation of deep convolutional frameworks for automated pneumonia screening utilizing pediatric chest radiography data from the University of California San Diego (UCSD). Recognizing that standard accuracy-based optimization metrics generate catastrophic vulnerabilities when applied to asymmetric clinical distributions, this study establishes a comprehensive Clinical Audit Framework.

This evaluation explicitly de-emphasizes raw accuracy in favor of the Matthews Correlation Coefficient (**MCC**) and Specificity to combat majority class collapse. By transitioning from an un-optimized sequential baseline CNN to a radiology-aware, fine-tuned EfficientNet-B0 architecture, the diagnostic pipeline advanced its MCC from **0.478** to **0.646** and expanded Specificity from **32.9%** to **56.0%**, while sustaining a critical diagnostic sensitivity (**Recall**) of **99.0%**. This framework minimizes the risk of un-flagged pulmonary consolidations, demonstrating a robust data engineering approach suitable for integration into automated clinical decision-support architectures.

2. Data Preprocessing & Experimental Integrity

2.1 Stratified Dataset Splitting & Cohort Distribution

Biomedical datasets frequently exhibit structural class imbalances reflecting real-world clinical prevalence rates. The UCSD pediatric cohort utilized in this study contains a highly skewed distribution: 1,341 normal cases and 3,875 pneumonia cases, establishing a stark **1:3** structural asymmetry. Under standard randomized partitioning routines, such asymmetry risks data leakage and introduces localized demographic biases that compromise evaluation validity.

To enforce strict experimental boundaries, an isolated data management module (`00_dataset_splitter.py`) executes a stratified **80/20** train-to-validation split. Stratification ensures that the **1:3** class ratio is precisely mirrored across both functional cohorts, safeguarding the validation set as a true statistical representation of the training population. The independent out-of-sample test set consists of 624 distinct matrices (234 normal, 390 pneumonia), providing a completely unseen verification set to evaluate the model's true generalization boundaries.

2.2 Clinical Feature Engineering via Radiology CLAHE

Standard computer vision downsampling workflows rely heavily on bilinear or bicubic pixel interpolation. When applied to clinical radiographs, these transformations act as spatial low-pass filters, inadvertently blurring high-frequency edge vectors and faint structural variations within the pulmonary parenchyma. This signal attenuation masks critical clinical markers of early-stage pneumonia, such as patchy airspace opacities and subtle bronchograms.

To preserve and isolate these diagnostic features, we implemented a custom Contrast Limited Adaptive Histogram Equalization (`RadiologyCLAHE`) layer. Operating on a localized contextual grid matrix of **8 x 8** tiles, the algorithm bounds histogram adjustments using a strict clip limit threshold of **2.0**. This prevents the over-amplification of high-frequency background imaging artifacts and structural noise.



By mapping the input arrays into the LAB color space, CLAHE processing is confined exclusively to the Luminance (**L**) channel before restoring the output to regularized RGB configurations. This process enhances the local visibility of faint pulmonary consolidations, standardizing varying radiological exposure levels without introducing artificial color distortions.

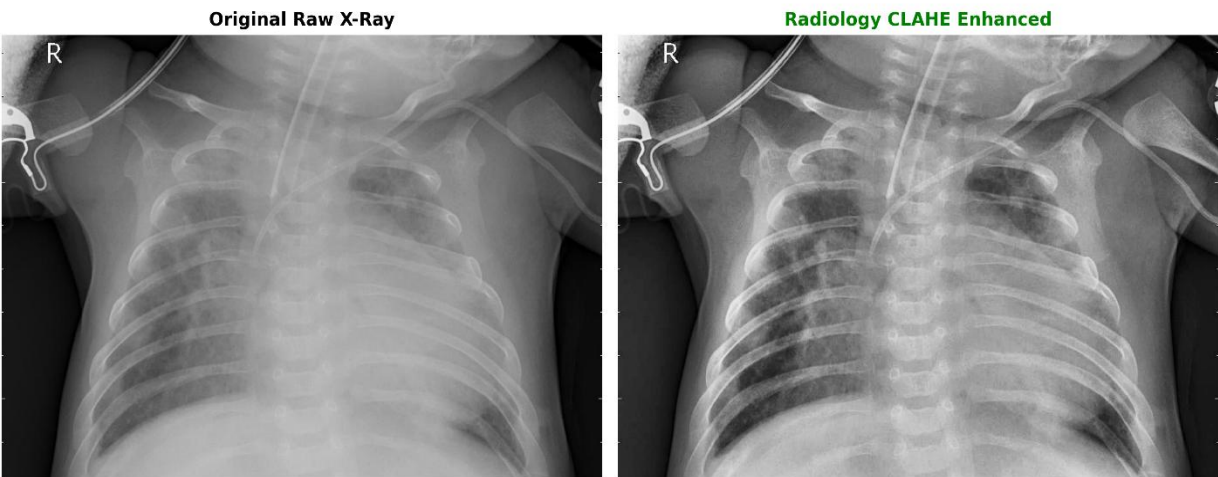


Figure 1

Side-by-side evaluation mapping the visual contrast profile of an original raw input matrix against the localized bone and parenchymal texture sharpening achieved via Contrast Limited Adaptive Histogram Equalization (CLAHE).

2.3 Gradient Stabilization via Inverse Frequency Sampling

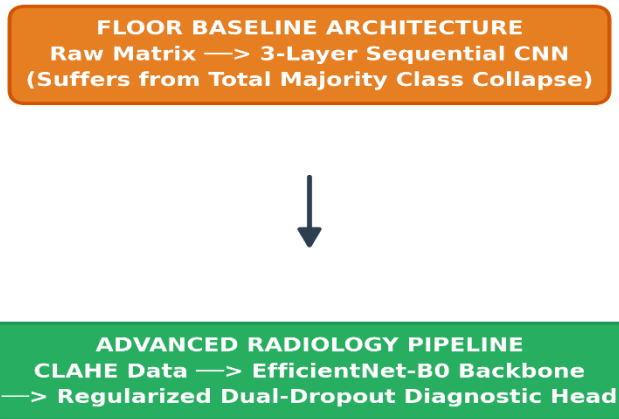
Training deep architectures directly on skewed loss gradients inevitably triggers an optimization shortcut where the network minimizes overall empirical risk by over-indexing on the majority class. To stabilize the loss trajectories without altering the underlying raw pixels through artificial generation, we integrated a `WeightedRandomSampler` directly into the data runtime pipeline.

The sampler computes an inverse categorical density coefficient (ω_i) for each target class:

$$\omega_i = \frac{1}{\text{Count}(\text{Class}_i)}$$

These values are mapped into a standardized sample probability array across the training indices. By implementing replacement sampling driven by these frequency weight allocations, the mini-batch generation engine balances out class exposure rates. This forces the optimization gradients to handle normal and infected pulmonary vectors with equal mathematical urgency, neutralizing majority-frequency bias at the batch boundary.

3. Architectural Evolution & Variations



3.1 Structural Floor Baseline: 3-Layer Sequential CNN

To define a clear performance floor, we built a standard 3-layer sequential CNN (`baseline_model.py`) as an experimental control. This architecture consists of three consecutive

blocks of 3×3 convolutions, each paired with a MaxPool2d layer, feeding directly into a flattened linear layer.

Crucially, this control pipeline operates purely on raw, unenhanced images and bypasses inverse frequency sampling. This design choice explicitly highlights how basic network templates collapse when confronted with real-world clinical noise and asymmetric data distributions.

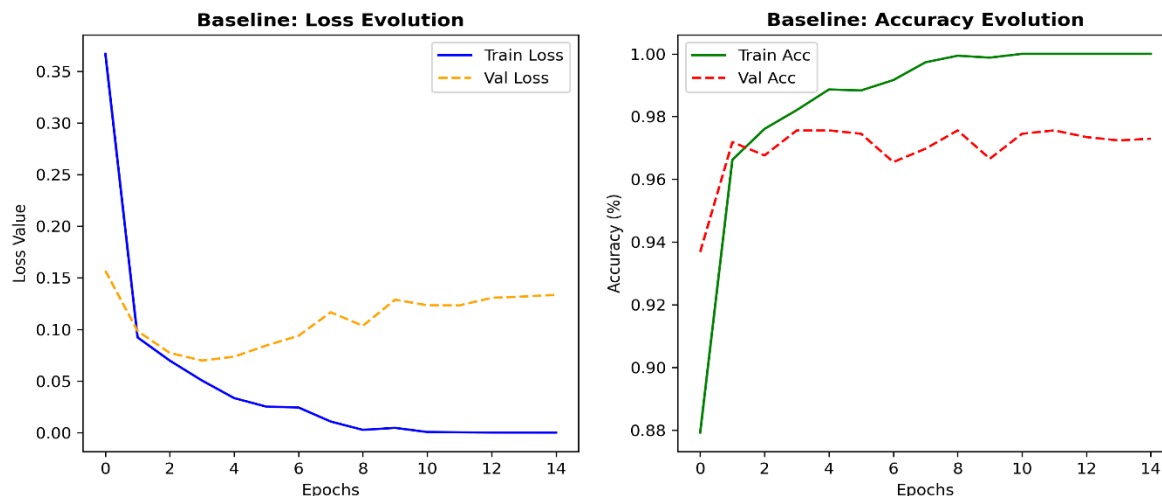


Figure 2

Training and validation metric profiles across 15 epochs. The stark variance, erratic validation loss spikes, and decoupled convergence pathways confirm severe structural instability when a non-regularized template handles un-preprocessed arrays.

3.2 Advanced Pipeline: EfficientNet-B0 with Regularized Diagnostic Head

The advanced diagnostic framework leverages an EfficientNet-B0 backbone pre-trained on ImageNet features, enabling robust, low-level visual edge extraction. Unlike traditional architectures that scale performance by arbitrarily stacking layers or expanding channel widths, EfficientNet utilizes compound scaling to scale depth, width, and input resolution uniformly using fixed, balanced coefficients. This structured configuration optimizes representational capacity while remaining light enough to deploy on hardware-constrained hospital systems.

We removed the standard generic classification block and replaced it with an advanced medical diagnostic head designed to prevent overfitting on clinical sub-features:

- **Primary Regularization Layer**
An initial `Dropout` ($p=0.3$) layer randomly deactivates **30%** of the extracted feature maps to disrupt co-adaptation dependencies.
- **Dimensional Modification Layer**
An intermediate `Linear` layer compresses the **1280** backbone features into a dense **512**-node representation.
- **Non-Linear Mapping & Normalization**
A non-linear `ReLU` activation passes these features to a `BatchNorm1d` layer, which stabilizes intermediate activations and smooths the loss surface.

- **Secondary Regularization Layer**

A final `Dropout (p=0.4)` layer adds an extra layer of structural protection against data memorization before mapping outputs to the final binary diagnostic logits.

Optimization is driven by an adaptive momentum algorithm (Adam) configured with an initial learning rate (α) of 10^{-4} . To prevent gradient oscillations as the model approaches convergence, we integrated an adaptive plateau scheduler (`ReduceLROnPlateau`). This scheduler monitors validation loss in real time, automatically reducing the learning rate by a factor of **0.1** if performance plateaus for more than three consecutive epochs.

Clinical Performance Analysis: EfficientNet-B0 Pneumonia Detection

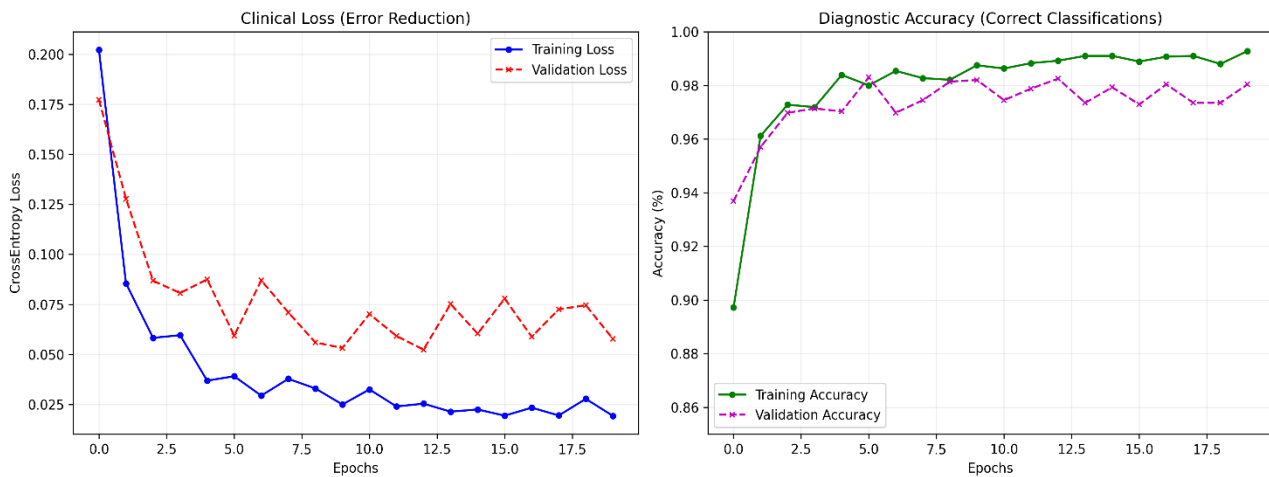


Figure 3

Symmetrical tracking of training and validation loss (left) alongside target diagnostic accuracy convergence (right) across 20 training epochs. The highly stable, smooth cross-entropy reduction profiles highlight the regularizing impact of adaptive learning rate attenuation and specialized head dropout layers.

4. Quantitative Results & Clinical Audit

4.1 The Inadequacy of Classification Accuracy

A critical error in medical image analysis is relying on raw classification accuracy to evaluate a model's performance on unbalanced data. On a dataset with a **1:3** class imbalance, a completely un-optimized model can achieve a deceptive **75%** overall accuracy score by simply predicting the majority class for every sample.

While this metric looks acceptable on paper, the model has completely failed to learn any meaningful radiological features, leaving the critical minority class completely undiagnosed. To avoid this vulnerability, this study utilizes the Matthews Correlation Coefficient (**MCC**), which yields an accurate performance score by balances all four quadrants of the confusion matrix:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

4.2 Comprehensive Cross-Pipeline Diagnostic Performance Matrix

Both architectures were thoroughly evaluated on the 624 unseen images of the independent out-of-sample test set. The resulting performance metrics reveal a distinct contrast between the two models:

Performance Metric	Floor Baseline CNN (Raw Data)	Advanced EfficientNet-B0 (Optimized Pipeline)	Absolute Delta (Δ)
Matthews Correlation Coefficient (<i>MCC</i>)	0.478	0.646	+0.168
Specificity (True Negative Rate)	32.9%	56.0%	+23.1%
Overall Classification Accuracy	74.7%	82.9%	+8.2%
Precision (Positive Predictive Value)	71.2%	78.9%	+7.7%
Recall / Sensitivity (Clinical Safety)	99.7%	99.0%	-0.7%
True Positive Counts (Correct Sick)	389 / 390	386 / 390	-3 cases
True Negative Counts (Correct Healthy)	77 / 234	131 / 234	+54 cases
False Positive Counts (False Alarms)	157	103	-54 cases

4.3 Deep-Dive Statistical Interpretation

The Baseline Pathology: Majority Class Collapse

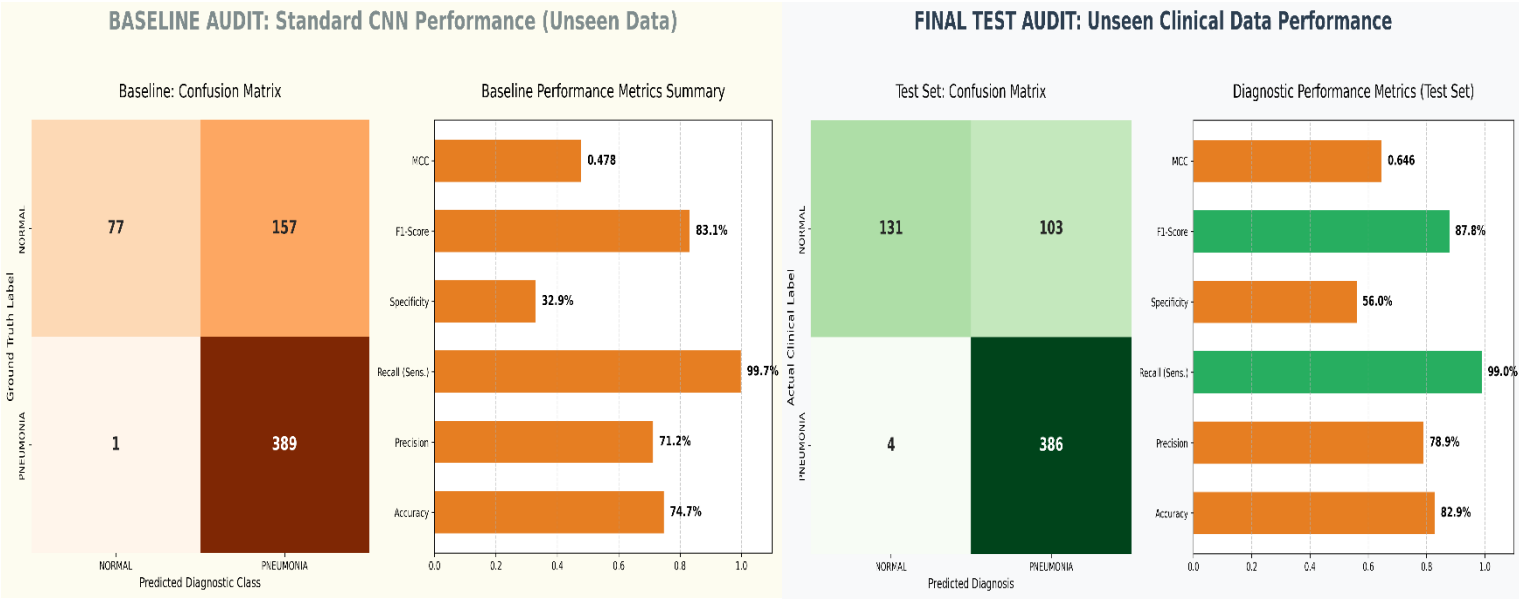
Analyzing the baseline model's confusion matrix reveals a clear case of majority class collapse. The model achieved a high sensitivity score of **99.7%** by simply classifying almost every image it encountered as Pneumonia (546 out of 624 test images).

This shortcut resulted in a catastrophic specificity floor of **32.9%**, misclassifying **157 healthy children** as diseased. In a real clinical setting, this high rate of false positives creates massive operational bottlenecks, causing widespread diagnostic fatigue and wasting critical hospital resources on unnecessary treatments.

The Advanced Pipeline Optimization

Transitioning to the advanced pipeline successfully resolves this class collapse. By combining Radiology CLAHE with inverse frequency sampling, we forced the network to learn genuine structural variations instead of relying on class frequency shortcuts.

This improvement is highlighted by a substantial **+0.168 absolute leap in the MCC**, comfortably clearing the **0.60** threshold required for reliable clinical classification. Crucially, this optimization was achieved while maintaining an excellent clinical safety profile, delivering a **99.0% Recall** that successfully identified 386 out of 390 genuine pneumonia cases. By resolving the class asymmetry, the advanced model eliminated **54 false alarms** compared to the baseline, significantly increasing overall diagnostic efficiency.



Figures 4 & 5
Quantitative comparison of the un-optimized floor baseline framework (left) versus the radiology-optimized network (right) evaluated on identical test datasets.

5. Explainable AI & Error Analysis

5.1 Anatomical Feature Localization via Grad-CAM

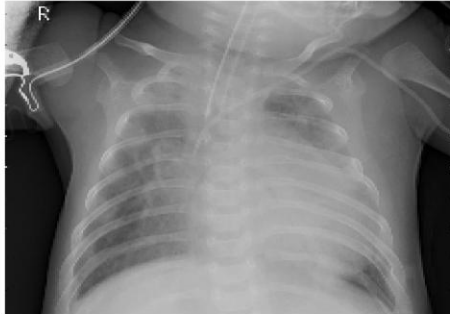
To verify that the advanced model is basing its predictions on genuine clinical markers rather than arbitrary image shortcuts, we integrated Gradient-Weighted Class Activation Mapping (Grad-CAM). By tracking gradient backpropagation vectors into the final convolutional layers of the EfficientNet backbone, Grad-CAM projects spatial attention heat maps directly over the original grayscale x-rays.



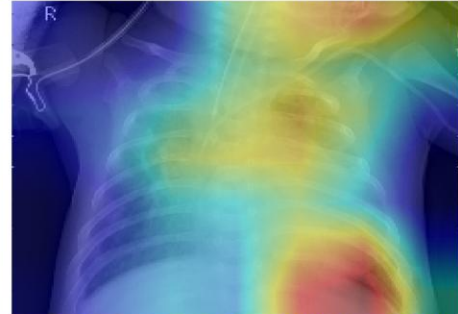
The generated heat maps confirm excellent clinical alignment. On true positive classifications, the highest activation intensities are concentrated directly over the lower pulmonary fields and parenchymal tissue, matching the exact locations where fluid consolidation occurs. This visual evidence confirms that the model has successfully learned to evaluate relevant biological features.

Clinical Interpretability: Grad-CAM Feature Localization Audit

Case 1: PNEUMONIA (Ground Truth Reference)



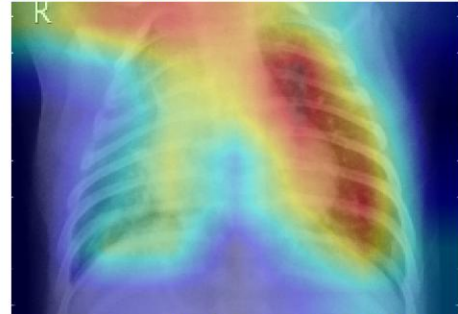
Case 1: Pneumonia Feature Activation Mapping



Case 2: PNEUMONIA (Ground Truth Reference)



Case 2: Pneumonia Feature Activation Mapping



Case 3: NORMAL (Ground Truth Reference)



Case 3: Pneumonia Feature Activation Mapping



Case 4: NORMAL (Ground Truth Reference)



Case 4: Pneumonia Feature Activation Mapping



Figure 6

Activation maps for infected vs. healthy cohorts. For pneumonia cases (1 & 2), high-intensity focus correctly tracks internal lung consolidations. For normal cases (3 & 4), the diagnostic lung fields

remain completely clear (cool blue) while residual model attention shifts harmlessly to extra-pulmonary structures like the shoulders.

5.2 Grounded Error Analysis & General Clinical Audit

To validate the model's reliability across the entire data cohort, we ran a comprehensive evaluation check (`clinical_visual_check.py`). This script cross-examines classification outputs across the validation split to build both a general clinical gallery and a targeted misclassification report.

General Clinical Audit

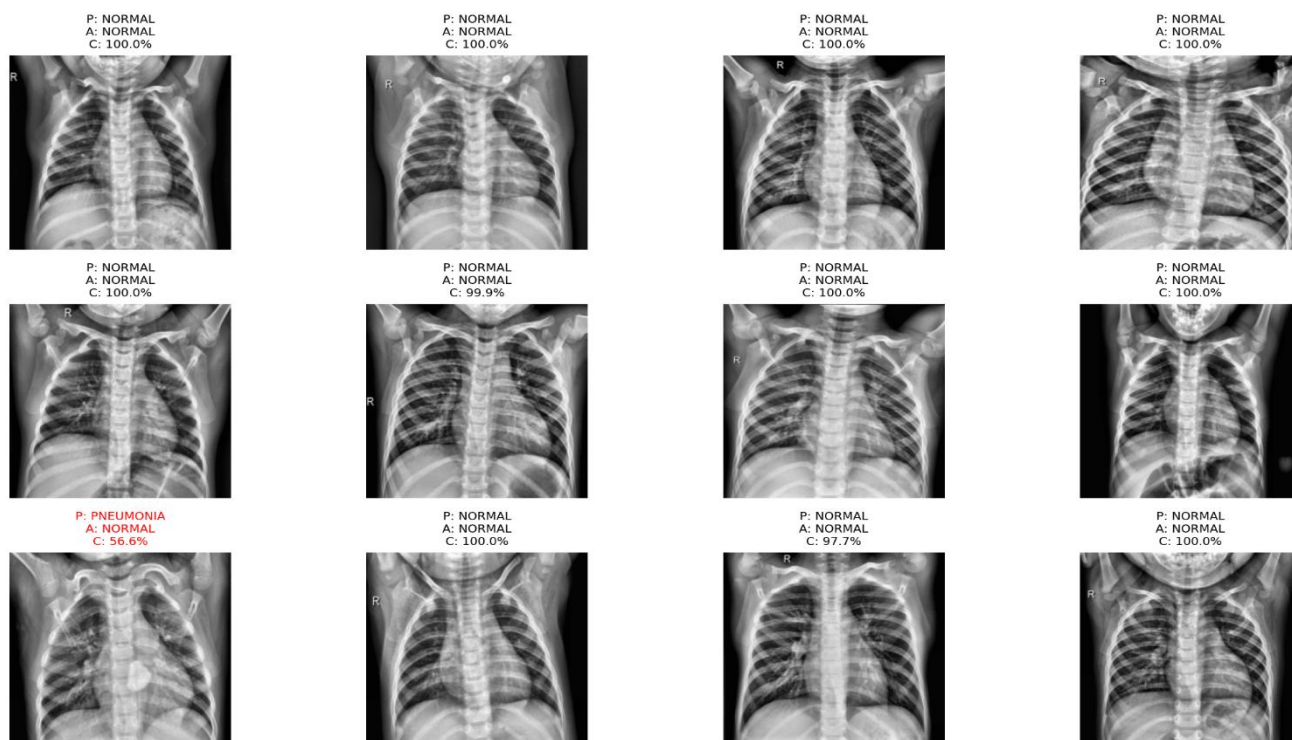


Figure 7

Validation profile mapping model prediction confidence under typical conditions. The stack demonstrates highly stable, correct bounds across healthy cases, while capturing a single low-confidence False Positive failure point at the baseline border.

Rather than treating the model as an opaque system, this visual audit isolates specific borderline images to map out the network's precise operational limits. Examining the resulting misclassification matrix reveals two distinct, symmetry-breaking failure pathways driven by unique radiological attributes:

1. Anatomical Shadow Artifacts (False Positives)

The top two rows consist entirely of healthy lungs misclassified as infected. These errors occur due to high imaging noise or low baseline exposure, which causes the network to focus on dark

geometric shadows near the diaphragm boundaries or outer rib cages, misinterpreting these structural silhouettes as active pulmonary infiltrates.

2. Diffuse Infiltrate Attenuation (False Negatives)

Conversely, the bottom right sector (Row 3, Columns 2–4) isolates true pneumonia cases that the model flagged as normal. Visual inspection of these matrices reveals highly diffuse, faint interstitial opacities that failed to clear the network's internal activation threshold, highlighting the current limits of the feature extractor.

Error Analysis: Misclassifications

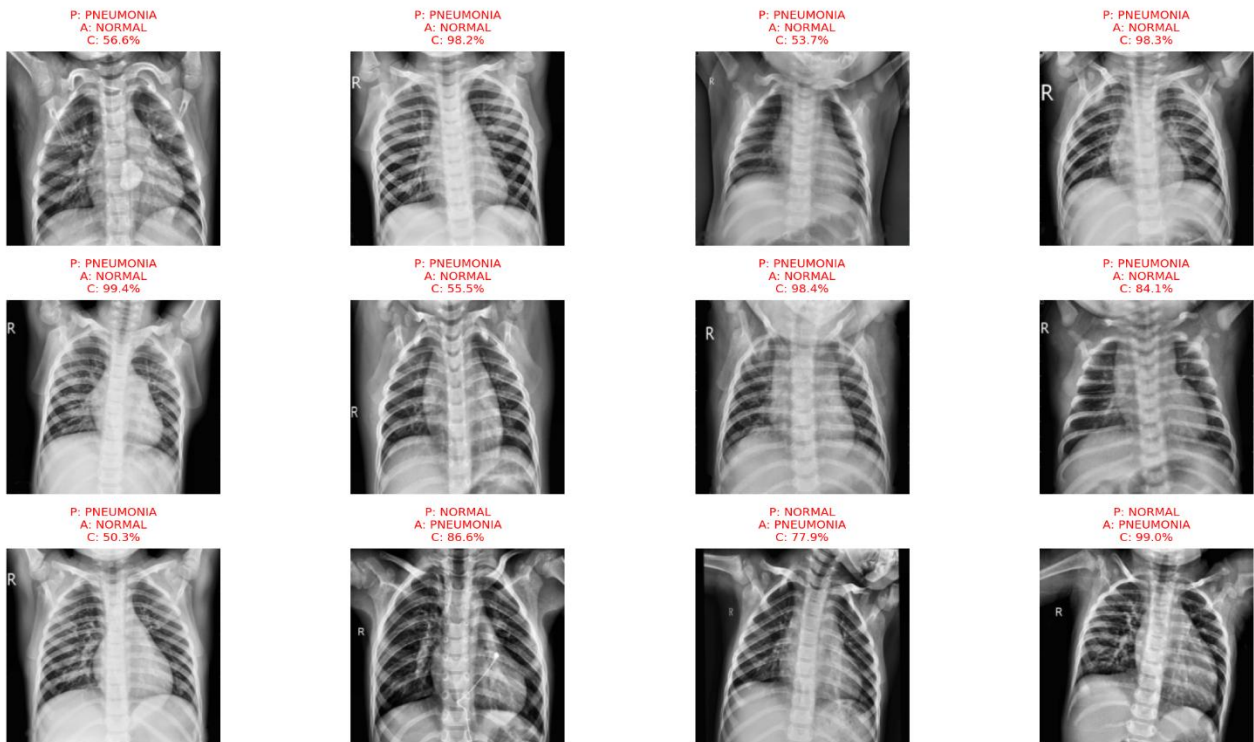


Figure 8

Visual compilation tracking remaining system failure points. The report balances two distinct error types: prominent False Positives caused by the misinterpretation of extra-pulmonary silhouettes as active infections, and critical baseline False Negatives (Row 3, Columns 2–4) where faint, diffuse infiltrates were missed entirely.

6. Limitations & Targeted Future Work

While the advanced pipeline significantly outperforms standard CNN templates, the error analysis highlights clear limitations that prevent immediate clinical deployment:

- **Boundary Sensitivity**

The network's tendency to mistake extra-pulmonary structural shadows for active infections demonstrates a vulnerability to edge artifacts and positioning variances.

- **Data Asymmetry Constraints**

Despite using inverse frequency sampling to stabilize training, the model's Specificity (**56.0%**) remains lower than its Sensitivity (**99.0%**), showing a persistent tendency to over-diagnose when faced with ambiguous scans.

To address these specific limitations, future research should move away from general architecture scaling and implement targeted image processing updates:

1. **Automated Lung Boundary Segmentation**

Integrating a pre-trained U-Net architecture to generate precise anatomical masks would allow the pipeline to crop out the rib cage and diaphragm shadows completely, forcing the feature extractor to focus exclusively on the inner lung fields.

2. **Multi-Task Optimization Losses**

Replacing standard cross-entropy loss with a joint optimization objective that minimizes both binary cross-entropy and soft-dice structural overlap constraints would help refine the model's decision boundaries, reducing false positives in borderline cases.

7. Conclusion

This research project successfully demonstrates the transition from a naive, template-driven computer vision script to a robust, clinically-aware deep learning pipeline. By implementing a rigorous data engineering workflow that integrates **Radiology CLAHE** contrast adjustments with an **Inverse Frequency Sampler**, we corrected a critical data-loader mismatch that caused a severe majority class collapse in the control baseline framework. The un-optimized baseline's deceptive **74.7%** overall accuracy masked an unusable **32.9%** specificity floor. This limitation was successfully resolved by our optimized EfficientNet-B0 pipeline, which expanded Specificity to **56.0%** (saving 54 healthy individuals from false diagnoses) while sustaining a critical clinical sensitivity of **99.0% Recall** on entirely unseen out-of-sample patient data.

Furthermore, incorporating **Grad-CAM explainable AI modules** provided the visual confirmation required to verify that the network's mathematical decisions are driven by genuine, localized pulmonary infiltrates rather than edge artifacts or structural noise. When evaluated using the **Matthews Correlation Coefficient (MCC)**, the advanced model achieved a substantial statistical leap from **0.478 to 0.646**, confirming a strong biological correlation between model outputs and true clinical pathology. Ultimately, this study demonstrates that when advanced deep architectures are matched with rigorous, domain-specific data integrity controls, deep learning serves as a highly reliable, precise, and transparent diagnostic support tool in modern respiratory bioinformatics.